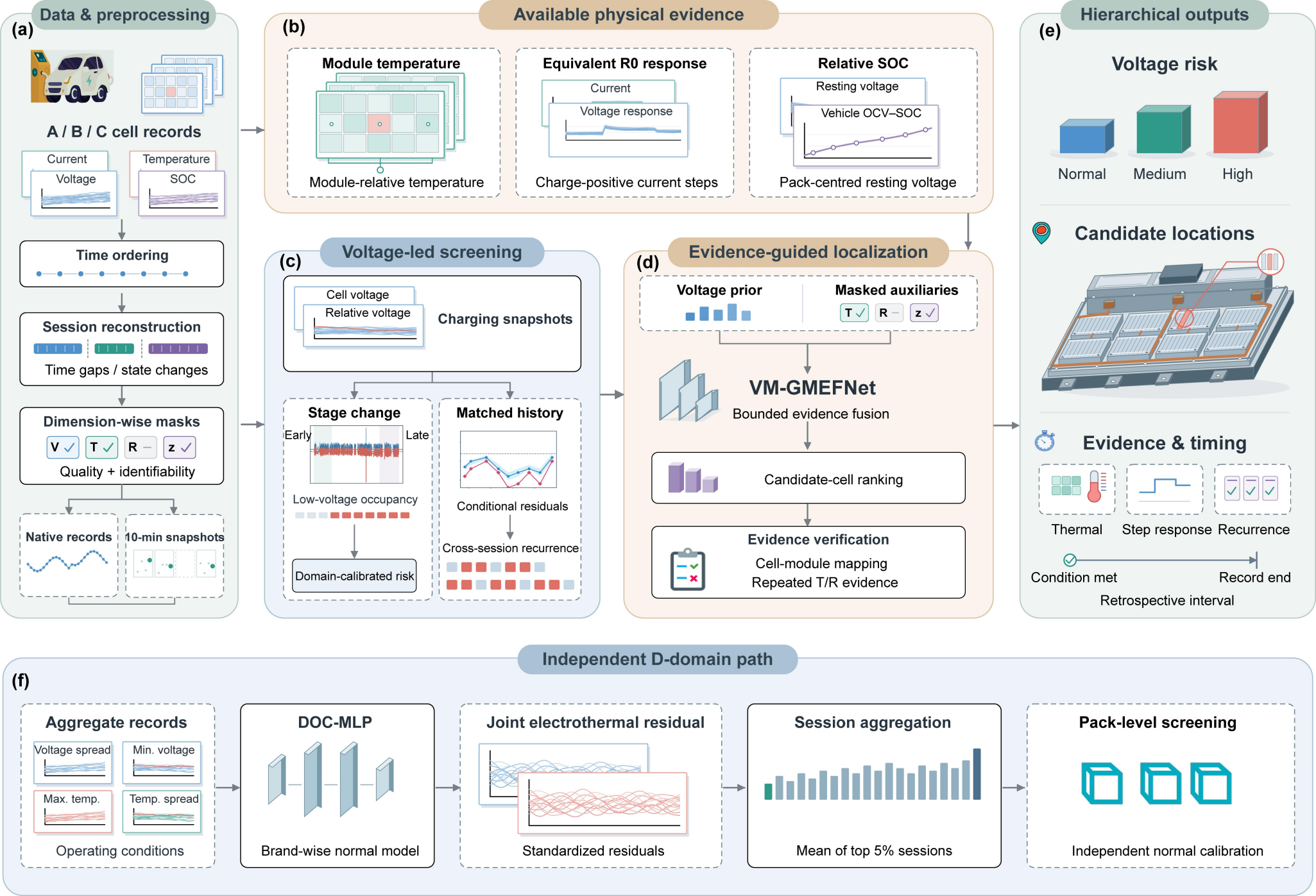

Data & preprocessing
(a)
A / B / C cell records
Current
Voltage
Temperature
SOC
Time ordering
Session reconstruction
Time gaps / state changes
Dimension-wise masks
V
T
R
z
Quality + identifiability
Native records
10-min snapshots
Available physical evidence
(b)
Module temperature
Module-relative temperature
Equivalent R0 response
Current
Voltage response
Charge-positive current steps
Relative SOC
Resting voltage
Vehicle OCV–SOC
Pack-centred resting voltage
Voltage-led screening
(c)
Cell voltage
Relative voltage
Charging snapshots
Stage change
Early
Late
Low-voltage occupancy
Domain-calibrated risk
Matched history
Conditional residuals
Cross-session recurrence
Evidence-guided localization
(d)
Voltage prior
Masked auxiliaries
T
R
z
VM-GMEFNet
Bounded evidence fusion
Candidate-cell ranking
Evidence verification
Cell-module mapping
Repeated T/R evidence
Hierarchical outputs
(e)
Voltage risk
Normal
Medium
High
Candidate locations
Evidence & timing
Thermal
Step response
Recurrence
Condition met
Record end
Retrospective interval
Independent D-domain path
(f)
Aggregate records
Voltage spread
Min. voltage
Max. temp.
Temp. spread
Operating conditions
DOC-MLP
Brand-wise normal model
Joint electrothermal residual
Standardized residuals
Session aggregation
Mean of top 5% sessions
Pack-level screening
Independent normal calibration